

**EXP:5****EDA – DATA VISUALIZATION****CODE:**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
data = {
    'Category': ['A', 'B', 'C', 'D'],
    'Values': [3, 7, 5, 4]
}
```

```
df = pd.DataFrame(data)
```

```
print("Dataset:")
```

```
display(df)
```

```
x = np.linspace(0, 10, 100)
```

```
y = np.sin(x)
```

```
plt.plot(x, y, label='Sine Wave')
```

```
plt.xlabel('X-axis')
```

```
plt.ylabel('Y-axis')
```

```
plt.title('Line Chart')
```

```
plt.legend()
```

```
plt.show()
```

```
plt.bar(df['Category'], df['Values'])
```

```
plt.xlabel('Category')
```

```
plt.ylabel('Values')
```

```
plt.title('Bar Chart')
```

```
plt.show()
```

```
random_data = np.random.randn(1000)
```

```
plt.hist(random_data, bins=20, edgecolor='black')
```

```
plt.xlabel('Value')
```

```
plt.ylabel('Frequency')
```

```
plt.title('Histogram')
```

```
plt.show()
```

## OUTPUT:

Dataset:

|   | Category | Values |
|---|----------|--------|
| 0 | A        | 3      |
| 1 | B        | 7      |
| 2 | C        | 5      |
| 3 | D        | 4      |



